

On the Number of Homogeneous Games Draft

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Abstract

The result we prove, if correct, would be a pretty cool contribution to the combinatorics of homogeneous games, as we show that the number of homogeneous games is related to Fishburn numbers[8, 6]. These numbers show up in many unrelated-looking settings typically in things with partially constrained and ordered structures. Due to this, a plethora of consequences ranging from asymptotics, bijections with other objects like linearized chord diagrams or interval orders, neat congruences, combinatorics of certain matrices, etc are now possible[2, 1, 4, 6]. It also seems that the number of homogeneous games with n players is currently only known up to $n = 9$, namely the terms being 0, 1, 3, 8, 23, 76, 293, 1307, 6642, 37882 (given in OEIS A189359[7]). Our contribution makes it easy to compute h_n for any n .

1 Statement of the result

Let h_n be the number of homogeneous games with n players, counted under the standard convention used in the enumeration of homogeneous games[3, 7]. As already mentioned in the abstract it seems that the number of homogeneous games h_n is currently only known up to $n = 9$, and our contribution makes it possible to compute effectively h_n for any n . We prove that for $n \geq 1$ we have

$$h_n - h_{n-1} = F_n$$

where F_n are the Fishburn numbers[8], whose sequence is 1, 1, 2, 5, 15, 53, ... Equivalently and quite nicely,

$$h_n = \sum_{j=1}^n F_j$$

Our proof uses work from Sudhölter, namely his recurrence for what are called reduced anti-step functions. Once the translation from homogeneous games to Sudhölter's functions is made, the rest of the argument is an ordinary generating-function calculation that doesn't require any game theory.

Let

$$F(x) = \sum_{n \geq 0} F_n x^n = \sum_{m \geq 0} \prod_{i=1}^m (1 - (1-x)^i)$$

be the Fishburn generating function, written in the standard form[6]. The main result is the following

Theorem 1.1. *For every $n \geq 1$,*

$$h_n - h_{n-1} = F_n$$

Equivalently,

$$h_n = \sum_{j=1}^n F_j$$

In generating-function form,

$$\sum_{n \geq 0} h_n x^n = \frac{F(x) - 1}{1 - x}$$

2 Prerequisites and Sudhölter's functions

Some context is given in this section. Not all of it is important, and the proof can be understood without most of this section, only important thing really is understanding the idea behind a 'dummy' player and the formulas of Sudhölter, so I will go over the prerequisite game theory stuff quickly. A simple game is a voting game where every coalition of players is either winning or losing, and adding more players to a winning coalition cannot make it losing. A weighted game is one that can be represented by weights and a quota, so that a coalition is winning exactly when its total weight is at least the quota. A weighted game is called homogeneous if it has a weighted representation in which every minimal winning coalition has total weight exactly equal to the quota. In other words, all minimal ways to win have the same total weight. A player is called a dummy if that player never changes the outcome of any coalition. Equivalently, adding a dummy player to a losing coalition never makes it winning. Dummies may be thought of as zero-weight players.

We write h_n for the number of homogeneous games with n players, under the standard equivalence convention, and g_n for the number of no-dummy homogeneous games with n players. Since every homogeneous game is obtained uniquely from a no-dummy homogeneous game by adjoining some number of final dummy players, Sudhölter's Lemma 1.1 gives[5]

$$h_n = \sum_{i=1}^n g_i$$

equivalently,

$$h_n - h_{n-1} = g_n$$

and thus it suffices to show that Theorem 1.1 is equivalent to proving

$$g_n = F_n$$

We use Sudhölter's enumeration of no-dummy homogeneous games, where he first reduces homogeneous games by removing dummies and one-person winning coalitions, then identifies the remaining games with incidence vectors and anti-step functions. His Lemma 1.1 gives the dummy deletion and the restoration of initial one-person-winning players, Lemma 1.4 gives the terminal-step deletion, Theorem 2.10 identifies homogeneous games without dummies and steps with incidence vectors, and Definition 3.1 identifies incidence vectors with anti-step functions[5].

Sudhölter then partitions the anti-step functions by type (k, p) in Definition 3.7[5]. We write $a_n^{k,p}$ for the number of n -person anti-step functions of type (k, p) , and

$$a_n^k := a_n^{k,0}$$

for the anti-step functions. The recurrence we use is stated below for these reduced functions.

For each $k \geq 1$,

$$A_k(x) = \sum_{n \geq 2k} a_n^k x^n$$

Sudhölter's formula gives[5]

$$a_n^{k,p} = a_{n-p}^k \binom{k+p-1}{p}$$

so summing over $p \geq 0$ gives the factor

$$\sum_{p \geq 0} \binom{k+p-1}{p} x^p = \frac{1}{(1-x)^k}$$

This is why in the next section we define

$$C_k(x) = \frac{A_k(x)}{(1-x)^k}$$

and

$$f(z) = \sum_{k \geq 1} C_k(x) z^k$$

so that

$$f(1) = \sum_{k \geq 1} C_k(x)$$

Thus $f(1)$ is the generating function on Sudhölter's anti-step side after the reduced counts have been expanded back to the full type (k, p) counts. There are two steps in translating this series to no-dummy homogeneous games. First, Sudhölter's terminal-step deletion, Lemma 1.4, shifts[5] the player number down by one. Thus a term counted by x^n in $f(1)$ first contributes to a no-dummy homogeneous game on $n - 1$ players. This accounts for the factor x^{-1} .

Second, the restoration of initial one-person-winning players, Lemma 1.1, says that we may adjoin any number of such players. If a game first contributes at size $n - 1$, then after adjoining $s \geq 0$ such players it contributes at sizes $n - 1, n, n + 1, \dots$ which contributes the geometric factor

$$1 + x + x^2 + \dots = \frac{1}{1-x}$$

Hence a term counted by x^n in $f(1)$ contributes

$$x^{n-1}(1 + x + x^2 + \dots) = \frac{x^{n-1}}{1-x}$$

to the no-dummy homogeneous-game generating function, and therefore, with

$$G(x) = \sum_{n \geq 1} g_n x^n$$

we get

$$G(x) = \frac{f(1)}{x(1-x)} \tag{1}$$

Remark 2.1. $a_2^1 = 1$ is included here under the shift by one player and the restoration of initial one-person-winning players, it contributes

$$\frac{x}{1-x}$$

to $G(x)$. This is the generating function for the 'exceptional no-dummy games' $(1; 1, \dots, 1)$, one in each positive number of players, though this game theory angle is NOT needed for our proof[5].

Remark 2.2. In Sudhölter's work and to tabulate the known values of h_n up until $n = 9$, incidence vectors are required for counting h_n , whereas with our method/proof this will not be needed[5].

3 The generating functions

We now introduce the generating functions attached to the reduced anti-step functions. For each $k \geq 1$, define

$$A_k(x) = \sum_{n \geq 2k} a_n^k x^n$$

The lower limit $n \geq 2k$ reflects the fact that $a_n^k = 0$ unless $n \geq 2k$. Define

$$C_k(x) = \frac{A_k(x)}{(1-x)^k}$$

and

$$f(z) = \sum_{k \geq 1} C_k(x) z^k$$

The variable z is only for bookkeeping, and in particular,

$$f(1) = \sum_{k \geq 1} C_k(x)$$

The reason we want to find $f(1)$ is that, by (1),

$$\frac{f(1)}{x(1-x)} = \sum_{n \geq 1} g_n x^n$$

we get h_n through

$$h_n = \sum_{i=1}^n g_i$$

4 The recurrence

The reduced numbers $a_n^k = a_n^{k,0}$ satisfy $a_n^k = 0$ unless $k \leq \lfloor \frac{n}{2} \rfloor$, and $a_2^1 = 1$ and $a_3^1 = 0$. Combining Sudhölter's Theorem 3.8 with Lemma 3.13 gives[5] the following recurrence for $n \geq 3$,

$$a_{n+1}^1 = \sum_{k=1}^{\lfloor n/2 \rfloor} \sum_{p=0}^{n-2k} \binom{k+p-1}{p} a_{n-p}^k \quad (2)$$

For $k \geq 2$, in the valid range

$$2 \leq k \leq \left\lfloor \frac{n+1}{2} \right\rfloor$$

the recurrence gives

$$a_{n+1}^k = \sum_{p=0}^{n-2k+2} \left[\sum_{j=k-1}^{\lfloor (n-p)/2 \rfloor} \binom{j}{k-1} \binom{j+p-1}{p} a_{n-p}^j \right] - a_n^{k-1} \quad (3)$$

So the goal is to turn these recurrences into a functional equation for $f(z)$ and solve that for $z = 1$.

5 From the recurrence to a functional equation

The $k = 1$ recurrence

Start with (2). Multiply both sides by x^{n+1} and sum over $n \geq 3$.

The left-hand side (I will just write ‘LHS’ from now on) is

$$\sum_{n \geq 3} a_{n+1}^1 x^{n+1}$$

Since

$$A_1(x) = \sum_{N \geq 2} a_N^1 x^N$$

the LHS sum is $A_1(x)$ with the terms $N = 2$ and $N = 3$ removed. Since $a_2^1 = 1$ and $a_3^1 = 0$, the LHS is simply

$$A_1(x) - x^2 - (0)x^3 = A_1(x) - x^2$$

Now let’s consider the RHS. The recurrence is only valid for $n \geq 3$, but if we look at beginning the sum at $n = 2$, the expression becomes much cleaner

$$x \sum_{k \geq 1} \frac{A_k(x)}{(1-x)^k} = xf(1)$$

however for $n = 2$ we have included one term that does not belong to the recurrence, so we have to account for this. When $n = 2$, the only possible contribution is $k = 1$ and $p = 0$ and it contributes $x^3 a_2^1 = x^3$. Therefore the actual RHS, summed over $n \geq 3$, is

$$xf(1) - x^3$$

thus putting it together

$$A_1 - x^2 = xf(1) - x^3$$

Equivalently,

$$A_1 = x^2(1-x) + xf(1)$$

Since $A_1 = (1-x)C_1$, this gives

$$(1-x)C_1 = x^2(1-x) + xf(1) \tag{4}$$

The $k \geq 2$ recurrences

Now fix $k \geq 2$. We multiply (3) by x^{n+1} and sum over $n \geq 2k-1$ which is the natural range because the LHS is a_{n+1}^k , and terms with k first occur when $n+1 \geq 2k$, so therefore the LHS sums exactly to

$$A_k(x)$$

now look at the RHS. In the main double sum, put

$$m = n - p$$

The factor a_{n-p}^j becomes a_m^j and for fixed j , summing over $p \geq 0$ produces the binomial series

$$\sum_{p \geq 0} \binom{j+p-1}{p} x^p = \frac{1}{(1-x)^j}$$

and thus the unrestricted main contribution would be

$$x \sum_{j \geq k-1} \binom{j}{k-1} \frac{A_j(x)}{(1-x)^j} = x \sum_{j \geq k-1} \binom{j}{k-1} C_j$$

However, this unrestricted sum includes one boundary term that is not allowed by the original range $n \geq 2k-1$ once again. The only forbidden case is when $j = k-1$ and $m = 2k-2$ and $p = 0$, which contributes $x^{2k-1} a_{2k-2}^{k-1}$. Now, lookin at the term $-a_n^{k-1}$, summing it over $n \geq 2k-1$ gives

$$-x \sum_{n \geq 2k-1} a_n^{k-1} x^n$$

But

$$A_{k-1}(x) = \sum_{n \geq 2k-2} a_n^{k-1} x^n$$

So, quite nicely we get

$$-x \sum_{n \geq 2k-1} a_n^{k-1} x^n = -x A_{k-1}(x) + x^{2k-1} a_{2k-2}^{k-1}$$

i.e the same term appears once from the unrestricted main sum and once from the subtraction term, with opposite signs and cancel out. Hence no correction remains, and we get

$$A_k = x \sum_{j \geq k-1} \binom{j}{k-1} C_j - x A_{k-1}$$

Using $A_k = (1-x)^k C_k$, this becomes

$$(1-x)^k C_k = x \sum_{j \geq k-1} \binom{j}{k-1} C_j - x(1-x)^{k-1} C_{k-1} \quad (5)$$

Combining all $k \geq 2$

We now multiply (5) by z^{k-1} and sum over $k \geq 2$. The purpose of z is only to combine the infinitely many equations into one. The LHS becomes

$$\sum_{k \geq 2} (1-x)^k C_k z^{k-1} = \frac{f((1-x)z) - (1-x)C_1 z}{z}$$

with the subtraction term which becomes

$$-x \sum_{k \geq 2} (1-x)^{k-1} C_{k-1} z^{k-1} = -x f((1-x)z)$$

For the binomial term, write $r = k-1$. Then $r \geq 1$, and

$$x \sum_{r \geq 1} z^r \sum_{j \geq r} \binom{j}{r} C_j = x \sum_{j \geq 1} C_j \sum_{r=1}^j \binom{j}{r} z^r$$

and since

$$\sum_{r=1}^j \binom{j}{r} z^r = (1+z)^j - 1$$

this is

$$x(f(1+z) - f(1))$$

and therefore

$$\frac{f((1-x)z) - (1-x)C_1z}{z} = x(f(1+z) - f(1)) - xf((1-x)z)$$

Now, using (4), namely

$$(1-x)C_1 = x^2(1-x) + xf(1)$$

the $f(1)$ terms cancel, and we obtain

$$(1+xz)f((1-x)z) = xzf(1+z) + x^2(1-x)z \quad (6)$$

Which is the desired functional equation.

6 Solving the functional equation

The functional equation

$$(1+xz)f((1-x)z) = xzf(1+z) + x^2(1-x)z$$

relates the value of f at $(1-x)z$ to the value of f at $1+z$. Since we ultimately need $f(1)$, we will choose z so that $(1-x)z = 1$, then repeat the same substitution again and again. To make this clean we can set $u = (1-x)z$, then $z = u/(1-x)$ and the functional equation becomes

$$f(u) = \frac{xu}{1-x+xu}f\left(1+\frac{u}{1-x}\right) + \frac{x^2(1-x)u}{1-x+xu} \quad (7)$$

Let's start with $u_0 = 1$, define

$$u_{r+1} = 1 + \frac{u_r}{1-x}$$

and then a direct induction gives

$$u_r = \frac{1 - (1-x)^{r+1}}{x(1-x)^r}$$

For our base case $r = 0$,

$$\frac{1 - (1-x)}{x} = 1$$

now assume the formula holds for r , then

$$u_{r+1} = 1 + \frac{1 - (1-x)^{r+1}}{x(1-x)^{r+1}} = \frac{x(1-x)^{r+1} + 1 - (1-x)^{r+1}}{x(1-x)^{r+1}} = \frac{1 - (1-x)^{r+2}}{x(1-x)^{r+1}}$$

and now we substitute $u = u_r$ into (7). Since

$$1 - x + xu_r = 1 - x + x \frac{1 - (1-x)^{r+1}}{x(1-x)^r} = 1 - x + \frac{1 - (1-x)^{r+1}}{(1-x)^r} = (1-x)^{-r}$$

we get

$$\frac{xu_r}{1-x+xu_r} = 1 - (1-x)^{r+1}$$

and then

$$\frac{x^2(1-x)u_r}{1-x+xu_r} = x(1-x)(1-(1-x)^{r+1})$$

thus

$$f(u_r) = (1 - (1-x)^{r+1})f(u_{r+1}) + x(1-x)(1 - (1-x)^{r+1}) \quad (8)$$

therefore if we iterate (8) from $r = 0$ to $r = N - 1$, we get

$$f(1) = x(1-x) \sum_{m=1}^N \prod_{i=1}^m (1 - (1-x)^i) + \left(\prod_{i=1}^N (1 - (1-x)^i) \right) f(u_N)$$

Each factor $1 - (1-x)^i$ has zero constant term, so it is divisible by x . Therefore

$$\prod_{i=1}^N (1 - (1-x)^i)$$

is divisible by x^N , so the final term disappears as $N \rightarrow \infty$. Also Note $f(u_N)$ is an ordinary formal power series in x , with no negative powers of x . This follows from

$$u_N = \frac{1 - (1-x)^{N+1}}{x(1-x)^N} = (1-x)^{-N}(1 + (1-x) + \cdots + (1-x)^N)$$

which is a formal power series in x . Also

$$C_k(x) = \frac{A_k(x)}{(1-x)^k}$$

starts in degree at least $2k$, because $A_k(x)$ starts with x^{2k} . Hence

$$f(u_N) = \sum_{k \geq 1} C_k(x) u_N^k$$

is a well-defined formal power series. Therefore the remainder

$$\left(\prod_{i=1}^N (1 - (1-x)^i) \right) f(u_N)$$

is divisible by x^N . For any fixed coefficient of x^d , this remainder contributes nothing once $N > d$. Thus, as a formal power series, the remainder vanishes in the limit.

Letting $N \rightarrow \infty$, we obtain

$$f(1) = x(1-x) \sum_{m \geq 1} \prod_{i=1}^m (1 - (1-x)^i)$$

Equivalently,

$$1 + \frac{f(1)}{x(1-x)} = \sum_{m \geq 0} \prod_{i=1}^m (1 - (1-x)^i) \quad (9)$$

7 Proof of the main theorem

The right-hand side of (9) is the Fishburn generating function $F(x)$. Therefore

$$1 + \frac{f(1)}{x(1-x)} = F(x)$$

By the normalization (1),

$$G(x) = \frac{f(1)}{x(1-x)}$$

Hence

$$1 + G(x) = F(x)$$

or equivalently

$$G(x) = F(x) - 1$$

equating coefficients gives

$$g_n = F_n$$

for $n \geq 1$. Finally, since

$$h_n = \sum_{t=1}^n g_t$$

so

$$h_n - h_{n-1} = g_n = F_n$$

Therefore

$$h_n - h_{n-1} = F_n$$

for $n \geq 1$, and equivalently

$$h_n = \sum_{j=1}^n F_j$$

This proves Theorem 1.1.

8 Consequences

These are immediate consequences/results from what I could find. These will just be stated without proof.

8.1 Congruences

One example,

$$h_{qn+r} \equiv h_{qn+r-1} \pmod{q}$$

whenever $F_{qn+r} \equiv 0 \pmod{q}$ [1, 4]

8.2 Asymptotics

$$h_n \sim \left(\frac{6}{\pi^2}\right)^n n! \sqrt{n} \frac{12\sqrt{3}}{\pi^{5/2}} e^{\pi^2/12}$$

and in particular,

$$\frac{h_{n+1}}{nh_n} \rightarrow \frac{6}{\pi^2}$$

[6]

8.3 Generating function

$$\sum_{n \geq 0} h_n x^n = \frac{1}{1-x} \left(\sum_{m \geq 0} \prod_{i=1}^m (1 - (1-x)^i) - 1 \right)$$

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